

BitHeteroNet: A Heterogeneous Network Benchmark for Enhanced Anomaly Detection in Bitcoin Transactions

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Abstract

The rapid proliferation of cryptocurrencies, particularly Bitcoin, has intensified the demand for effective detection of illicit activities such as ransomware payments and money laundering. However, existing approaches are often limited in capturing the complex structural properties inherent to Bitcoin's Unspent Transaction Output (UTXO) system, notably its heterogeneity and one-to-many transaction-address interactions. To address the absence of a comprehensive benchmark that accurately reflects UTXO-based transactional behavior, we introduce BitHeteroNet, a novel benchmark designed to support robust Bitcoin fraud detection. BitHeteroNet comprises detailed features and categorical labels for 822,000 wallet addresses and 202,000 transactions, along with their interconnections, enabling a holistic representation of transaction-address relationships. This benchmark provides a foundational resource for developing and evaluating detection models that account for the intricate interdependencies within the Bitcoin ecosystem. Our benchmark and baseline implementations are available [here](#).

CCS Concepts

• **Mathematics of computing** → **Graph algorithms**; • **Computing methodologies** → **Neural networks**; **Anomaly detection**.

Keywords

Bitcoin, Cryptocurrency, Fraud Detection

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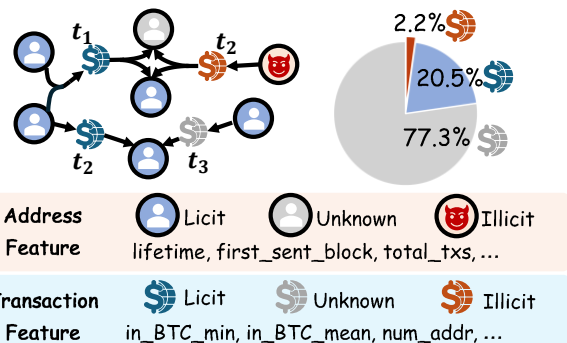


Figure 1: In the UTXO system (e.g., Bitcoin), transactions involve multiple wallet addresses and have unique features, such as time stamps. In our dataset, BitHeteroNet, most transactions are unclassified, while identified transactions are mostly licit, with only a small portion being illicit.

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1 Introduction

The rise of cryptocurrencies has created a decentralized financial ecosystem, enabling direct transactions without intermediaries. This growth has prompted traditional financial institutions to support crypto-fiat conversions, necessitating robust anomaly detection to identify illicit activities like money laundering and ransomware payments [1]. As one of the most widely adopted cryptocurrencies, Bitcoin has been the subject of extensive research in financial fraud detection. In particular, the introduction of the Elliptic [13] and Elliptic++ [3] datasets has enabled the application of Graph Neural Networks (GNNs) to extract meaningful node representations that facilitate fraud detection [5–7, 15].

However, existing dataset constructions have largely overlooked the most fundamental transaction model of Bitcoin, known as the Unspent Transaction Output (UTXO), as illustrated in Figure 1. The UTXO system's inherent complexity presents unique challenges for fraud detection. First, wallet addresses and transactions represent distinct data sources with heterogeneous attributes. Second, unlike

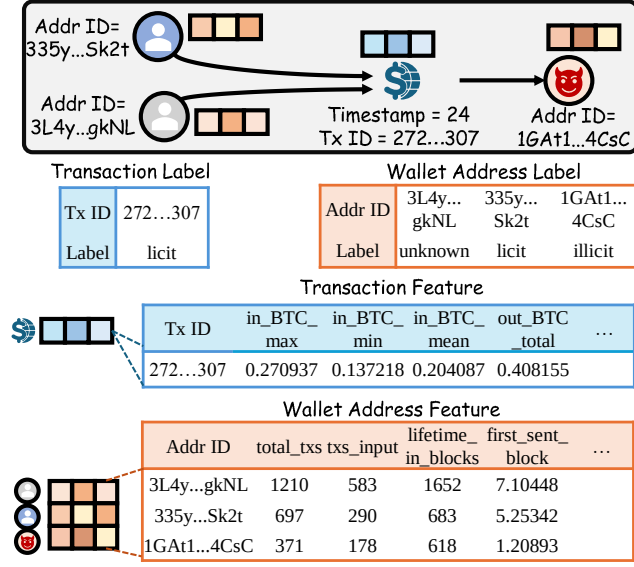


Figure 2: A toy example for a transaction and relevant wallet addresses as well as their features in the BitHeteroNet.

account-based systems (e.g., Ethereum) where transactions involve exactly one input and one output account, UTXO transactions can engage more than two addresses, thereby obscuring clear input-output relationships [8]. The Elliptic dataset [13] consists solely of Bitcoin transactions without wallet addresses, and Elliptic++ dataset [3] divides addresses and transactions into two separate subdatasets without their interdependencies.

Consequently, forensic analysis utilizing these datasets encounters a significant limitation: when a model predicts illicit transactions or addresses, the dependencies between transactions and addresses are either absent (Elliptic) or not examined in an integrated manner (Elliptic++). To fill this gap and promote the development of advanced UTXO modeling methods that can fully consider these dependencies, our study introduces the *BitHeteroNet* dataset, which enables researchers to perform fraud detection analyses on transactions and addresses within the UTXO system jointly. The difference of BitHeteroNet with existing Bitcoin datasets is shown in Table 1.

2 BitHeteroNet Benchmark

The BitHeteroNet is a Bitcoin transaction network encompassing 822,942 wallet addresses and 202,804 Bitcoin transactions as well as their features and interrelationships. The statistics are shown in Table 2. To facilitate comprehension, Figure 2 presents a simplified illustrative example of the information contained within our BitHeteroNet. The benchmark contains a wealth of features and labels for both transactions and wallet addresses. For transactions, we include a unique Tx ID and a label indicating the transaction is licit, illicit or unknown. The transaction features provided detailed information about Bitcoin flows, including the maximum, minimum and mean input amounts. These metrics offer insights into the scale and nature of each transaction. Wallet addresses are similarly detailed, with each assigned a unique Addr ID and a label classifying it as unknown, licit or illicit. The wallet address features capture the history and activity of each address, including the total number of transactions and the address's lifetime measured in

Table 1: The comparison of BitHeteroNet with existing Bitcoin fraud detection dataset.

Dataset	Elliptic [13]	Elliptic++ [3]	BitHeteroNet (Ours)
Wallet	×	✓	✓
Transaction	✓	✓	✓
Relationship	×	×	✓

Table 2: The statistics of the BitHeteroNet benchmark.

# Transaction	202,804
licit: illicit: unknown	4,545: 41,500: 156,759
# Wallet address	822,942
licit: illicit: unknown	14,266: 250,998: 557,678
# Transaction feature	183
# Wallet address feature	54
# Timestamps	49

Table 3: Some wallet address features and descriptions. (S) represents the feature as a single value, while (M) represents the feature has 5 values: total, min, max, mean, and median.

Feature	Description
BTC-transacted (M)	Total BTC transacted (sent+received)
BTC-sent (M)	Total BTC transacted (sent)
BTC-received (M)	Total BTC transacted (received)
Fees (M)	Total fees in BTC
Fees-share (M)	Total fees as share of BTC transacted
Txs-total (S)	Total number of blockchain transactions
Lifetime (S)	Lifetime in blocks
Block-first (S)	Block height first transacted in
Block-last (S)	Block height last transacted in
Block-first sent (S)	Block height first sent in
Block-first receive (S)	Block height first received in
...	...

blocks. This information allows for a comprehensive understanding of each address's behavior over time. Some wallet features are listed in Table 3. The benchmark also captures the relational structure of transactions, illustrating how different wallet addresses are connected through a single transaction. The labels are sourced from the Elliptic and Elliptic++ datasets, ensuring consistency and reliability in classification. Figure 2 top shows fund flow between addresses, complete with a timestamp and transaction ID. This structural information is crucial for understanding the dynamics of cryptocurrency transactions and potentially identifying illicit activities based on transactional patterns and wallet behaviors. Table 4 shows the storage files and respective descriptions.

3 Problem Definition

In this section, we first define the inputs of UTXO system, and then delineate the fraud detection tasks on wallet addresses and transaction, respectively.

Definition 3.1 (UTXO system). A UTXO system's input is denoted as $\mathbb{U} = \{\mathcal{X}^{\mathcal{W}}, \mathcal{X}^{\mathcal{B}}, \mathcal{R}^{(in)}, \mathcal{R}^{(out)}, g\}$, where $\mathcal{X}^{\mathcal{W}}$ and $\mathcal{X}^{\mathcal{B}}$ denote the feature sets of the address set $\mathcal{W}$ and the transaction set $\mathcal{B}$,

Table 4: The storage files for saving the information of BitHeteroNet. The *.npz files are loaded through the Numpy package. The data.pk1 is a dictionary with multiple keys saved in pickle format.

Type	File	Description
Feature & Label	address_attrs.npz	The features of wallet addresses
	addr_classes.npz	The labels (0: licit, 1: illicit, 2: unknown) of wallet addresses
	tx_features.npz	The features of transactions
	tx_time_step.npz	The occurrence timestamp of each transaction
	tx_classes.npz	The labels (0: licit, 1: illicit, 2: unknown) of transactions
Relational Structure	data.pk1(['Addr2Addr_id'])	A dictionary mapping from raw wallet address ID to an index
	data.pk1(['Tx2Tx_id'])	A dictionary mapping from raw transaction ID to an index
	data.pk1(['Tx_id2Output_Addr_ids'])	A dictionary mapping from transaction to the output addresses
	data.pk1(['Tx_id2Input_Addr_ids'])	A dictionary mapping from transaction to the input addresses

respectively. The set $\mathcal{R}^{(in)}$ defines the input relations, such that a pair $(w, b) \in \mathcal{R}^{(in)}$ signifies that address w is one of the input addresses of transaction b . The output relation set $\mathcal{R}^{(out)}$ is defined analogously. Additionally, for each transaction $b \in \mathcal{B}$, the function $g(b)$ returns the timestamp at which b occurs.

Remark 3.2 (The applications of UTXO). The UTXO transaction pattern is widely adopted in cryptocurrencies like Bitcoin, Litecoin, Zcash, and Dash¹. Given Bitcoin’s widespread adoption among cryptocurrencies, this study will utilize Bitcoin as a representative instance for examining the UTXO transaction system.

Leveraging the illicit, licit, and unknown labels of addresses and transactions in the *BitHeteroNet*, our study focuses on two fundamental tasks: *Illicit Address Detection* and *Illicit Transaction Detection*. The definitions of these tasks are as follows:

Definition 3.3 (Illicit Address Detection). Given UTXO system’s input $\mathbb{U}$ and a training dataset $\mathcal{D}^{\mathcal{W}} = \{(w, y_w) | w \in \mathcal{W}', y_w \in \{0, 1\}\}$, where $\mathcal{W}' \subseteq \mathcal{W}$ is a subset of addresses with known labels, 0 indicates a licit address, and 1 indicates an illicit address, the objective of the task is to train a model capable of identifying the legitimacy of unlabeled addresses.

Definition 3.4 (Illicit Transaction Detection). Given UTXO system’s input $\mathbb{U}$ and a training dataset $\mathcal{D}^{\mathcal{B}} = \{(b, y_b) | b \in \mathcal{B}', y_b \in \{0, 1\}\}$, where $\mathcal{B}' \subseteq \mathcal{B}$ is a subset of labeled transactions, 1 denotes an illicit transaction, and 0 denotes a licit transaction, the aim is to identify the legitimacy of unlabeled transactions.

4 Experiments

In this section, we experimentally demonstrate that utilizing our proposed dataset, *BitHeteroNet*, enables a more comprehensive modeling of the UTXO transaction mechanism in Bitcoin, thereby significantly enhancing the efficiency of fraud detection.

Task Settings. Across all tasks, we allocate 20% of labeled samples (both licit and illicit) for training, with the remaining samples divided in a 1:2 ratio between validation and test sets. For the illicit address detection task, we evaluate our method under both *transductive* and *inductive* settings. In the transductive setting, all addresses are incorporated into a single graph, whereas in the inductive setting, distinct subsets of nodes as well as their topologies are

visible only during their respective model stages. This comparative design is intended to assess the robustness of the model, particularly in scenarios involving missing or unobserved transactions between addresses. For the illicit transaction task, we employ two evaluation settings: *Random* and *Time-based*. In the random setting, transactions are arbitrarily partitioned. In the time-based setting, transactions are divided according to timestamps, with the earliest, intermediate, most recent transactions forming the training, validation, test sets, respectively. We assess the detection performance using three widely-accepted metrics: AUROC, AUPR and FPR@95.

Baselines. To evaluate the efficacy of *BitHeteroNet* on modeling the UTXO transaction patterns by incorporating the wallet addresses and transactions into a single heterogeneous graph, we evaluate the baselines through following steps. (i) We construct two homogeneous graphs for addresses and transactions, respectively. For instance, an edge is established between two addresses in the address graph if they have previously participated in a common transaction. (ii) We employ a homogeneous GNN for the graph, tune its parameters on the training node set, and evaluate the model’s performance on the validation and test sets. The baselines include (i) General GNN detectors, e.g., GCN [4] and GAT [11], (ii) fraud detection GNN, BWGNN [10] and PMP [16]. We also include MLP as a single baseline solely based on the features.

A Simple Heterogeneous GNN for Bitcoin. We implement a simple heterogeneous GAT for Bitcoin Transaction Network (Bit-GAT), which adopts GAT as the backbone, for *BitHeteroNet* to illustrate the effectiveness of our joint modeling paradigm. We first use two different linear layers to transform the features of wallet addresses and transactions into the same dimension. To adapt the heterogeneous nodes, we modify the attention of GAT from wallet w to transaction b as following:

$$\alpha_{wb} = \frac{\exp\left(\mathbf{a}^T [\mathbf{W}\mathbf{h}_w \parallel \mathbf{W}\mathbf{h}_b \parallel \mathbf{W}_r \mathbf{r}_{\psi(\langle w, b \rangle)}]\right)}{\sum_{k \in \mathcal{N}_w} \exp\left(\mathbf{a}^T [\mathbf{W}\mathbf{h}_w \parallel \mathbf{W}\mathbf{h}_k \parallel \mathbf{W}_r \mathbf{r}_{\psi(\langle w, k \rangle)}]\right)}, \quad (1)$$

where $\mathbf{W}_r$ is a learnable matrix to transform embedding according to edge types, i.e., {in, out}, and $\mathcal{N}_w$ is the transaction set involving wallet w . The α_{bw} is defined similarly to α_{wb} .

Implementations. We optimize all baseline models and our proposed models using a two-layer architecture. The hyperparameter tuning process involved selecting the learning rate from {0.01, 0.005,

¹<https://explorer.bitquery.io/platform/bitcoin>

Table 5: Illicit address and transaction detection performance across various task settings on the BitHeteroNet dataset.

Task \ Method		Illicit Address Detection						Illicit Transaction Detection					
		Transductive			Inductive			Random			Time-based		
		AUROC $\uparrow$			AUPR $\uparrow$			FPR@95 $\downarrow$					
Disjunct Model	MLP	78.52	15.86	88.51	68.67	12.24	87.80	60.88	12.04	96.71	56.57	11.16	97.45
	GCN	79.51	18.10	70.91	67.24	15.89	74.05	80.70	30.48	68.89	67.16	16.44	93.40
	GAT	81.86	26.17	60.05	65.23	17.18	71.37	83.59	47.66	47.95	72.81	38.13	60.30
	BWGNN	76.50	22.78	61.22	67.68	15.38	77.08	82.56	38.97	57.33	74.86	34.28	64.70
	PMP	84.96	42.73	49.24	71.16	21.63	61.73	85.38	44.89	45.43	71.19	30.82	69.26
Joint Model	Bit-GAT	88.73	63.80	38.17	74.52	34.59	52.31	91.68	76.56	29.49	77.84	41.89	58.75

0.001}, utilizing Adam optimizer over 50 epochs, and choosing hidden state dimension d from {32, 64, 128}. To ensure the robustness of our results, we conduct five independent runs for each dataset and report the average performance.

Results. The illicit address and transaction detection results are shown in Table 5. Overall, Bit-GAT shows superior effectiveness and stability across all evaluation metrics in all tasks, demonstrating effective integration of heterogeneous transaction features and address features. For instance, in the transductive setting, Bit-GAT surpasses PMP with absolute improvements of 3.8% and 21.1% in AUROC and AUPR respectively, while reducing FPR@95 by 11.1%.

5 Related Work

The emergence of blockchain technologies such as Bitcoin and Ethereum has introduced significant cybersecurity challenges, especially in the area of fraud detection. The global accessibility and pseudo-anonymity provided by blockchain platforms have facilitated the misuse of cryptocurrencies in various criminal activities—including ransomware attacks and money laundering, which are now recognized as among the fastest-growing forms of cyber-crime [9]. Since the release of the Elliptic dataset [13], considerable research has been devoted to fraud detection within Bitcoin transaction networks. For instance, GuiltyWalker [7] derives new features by examining transaction-to-transaction graph structures and measuring distances to known illicit transactions. Inspection-L [6] employs a Graph Isomorphism Network [14] as its transaction encoder and utilizes Deep Graph Infomax [12] as a pre-training strategy. In a similar vein, Alarab and Prakoonwit [2] combine a graph-based LSTM with a GCN [4] to identify illicit transactions and integrate an active learning framework to mitigate the scarcity of labeled illicit data. Moreover, the Elliptic++ dataset [3] expands upon Elliptic by incorporating relevant Bitcoin addresses in a separate subdataset. Despite these advances, most existing studies remain centered on applying conventional GNNs to learn transaction representations from transaction-to-transaction graphs [2, 6].

6 Conclusion

In this study, we introduced BitHeteroNet, a Bitcoin Transaction Dataset, encompassing a multitude of Bitcoin transactions and wallet addresses, their associated features and labels, as well as their interdependencies. Our empirical evaluations demonstrated that our joint modeling method tailored for UTXO transactions significantly outperforms leading baselines in all scenarios.

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References

- [1] Aisha Abdallah, Mohd Aizaini Maarof, and Anazida Zainal. 2016. Fraud detection system: A survey. *Journal of Network and Computer Applications* (2016), 90–113.
- [2] Ismail Alarab and Simant Prakoonwit. 2023. Graph-based lstm for anti-money laundering: Experimenting temporal graph convolutional network with bitcoin data. *Neural Processing Letters* (2023), 689–707.
- [3] Youssef Elmougy and Ling Liu. 2023. Demystifying Fraudulent Transactions and Illicit Nodes in the Bitcoin Network for Financial Forensics. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 3979–3990.
- [4] Thomas N Kipf and Max Welling. 2016. Semi-Supervised Classification with Graph Convolutional Networks. In *Proc. of ICLR*.
- [5] Sijia Li, Gaopeng Gou, Chang Liu, Chengshang Hou, Zhenzhen Li, and Gang Xiong. 2022. TTAGN: Temporal transaction aggregation graph network for ethereum phishing scams detection. In *Proceedings of the ACM Web Conference 2022*. 661–669.
- [6] Wai Weng Lo, Gayan K Kulatilleke, Mohanad Sarhan, Siamak Layeghy, and Marius Portmann. 2023. Inspection-L: self-supervised GNN node embeddings for money laundering detection in bitcoin. *Applied Intelligence* (2023), 19406–19417.
- [7] Catarina Oliveira, João Torres, Maria Inês Silva, David Aparício, João Tiago Ascensão, and Pedro Bizarro. 2021. GuiltyWalker: Distance to illicit nodes in the Bitcoin network. *arXiv preprint arXiv:2102.05373* (2021).
- [8] Yuxin Qi, Jun Wu, Hansong Xu, and Mohsen Guizani. 2023. Blockchain Data Mining With Graph Learning: A Survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2023).
- [9] Kiarash Shamsi, Friedhelm Victor, Murat Kantarcioglu, Yulia Gel, and Cuneyt G Akcora. 2022. Chartalist: Labeled graph datasets for utxo and account-based blockchains. *Proc. of NeurIPS* (2022), 34926–34939.
- [10] Jianheng Tang, Jiajin Li, Ziqi Gao, and Jia Li. 2022. Rethinking graph neural networks for anomaly detection. In *Proc. of ICML*. 21076–21089.
- [11] Petar Velicković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. 2018. Graph Attention Networks. In *Proc. of ICLR*.
- [12] Petar Velickovic, William Fedus, William L Hamilton, Pietro Liò, Yoshua Bengio, and R Devon Hjelm. 2019. Deep graph infomax. *ICLR (Poster)* (2019), 4.
- [13] Mark Weber, Giacomo Domeniconi, Jie Chen, Daniel Karl I Weidele, Claudio Bellei, Tom Robinson, and Charles E Leiserson. 2019. Anti-money laundering in bitcoin: Experimenting with graph convolutional networks for financial forensics. *arXiv preprint arXiv:1908.02591* (2019).
- [14] Keyulu Xu, Weihua Hu, Jure Leskovec, and Stefanie Jegelka. 2019. How Powerful are Graph Neural Networks?. In *Proc. of ICLR*.
- [15] Jiajun Zhou, Chenkai Hu, Jianlei Chi, Jiajing Wu, Meng Shen, and Qi Xuan. 2022. Behavior-aware account de-anonymization on ethereum interaction graph. *IEEE Transactions on Information Forensics and Security* (2022), 3433–3448.
- [16] Wei Zhuo, Zemin Liu, Bryan Hooi, Bingsheng He, Guang Tan, Rizal Fathony, and Jia Chen. 2024. Partitioning Message Passing for Graph Fraud Detection. In *The Twelfth International Conference on Learning Representations*.